

RNA-Seq Analysis

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Setup

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

```
findMatches
```

The following objects are masked from 'package:base':

```
expand.grid, I, unname
```

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

```
colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

Differential Expression Analysis

```
metaFile <- "data/GSE37704_metadata.csv"
countFile <- "data/GSE37704_featurecounts.csv"

colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
      condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
      length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
      SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Q1:

```
countData <- as.matrix(countData[, -1])
head(countData)
```

```
      SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
```

ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q2:

```
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

Running DESeq2

```
dim(countData)
```

```
[1] 15975      6
```

```
dim(colData)
```

```
[1] 6 1
```

```
countData <- read.csv(countFile, row.names=1)
countData <- countData[,-1]
countData <- as.matrix(countData)

colData <- read.csv(metaFile, row.names=1)

all(colnames(countData) == rownames(colData))
```

```
[1] TRUE
```

```
dds <- DESeqDataSetFromMatrix(countData = countData,  
colData = colData,  
design = ~ condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet  
dim: 19808 6  
metadata(1): version  
assays(4): counts mu H cooks  
rownames(19808): ENSG00000186092 ENSG00000279928 ... ENSG00000277475  
                  ENSG00000268674  
rowData names(22): baseMean baseVar ... deviance maxCooks  
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371  
colData names(2): condition sizeFactor
```

```
res = results(dds)
```

```
summary(res)
```

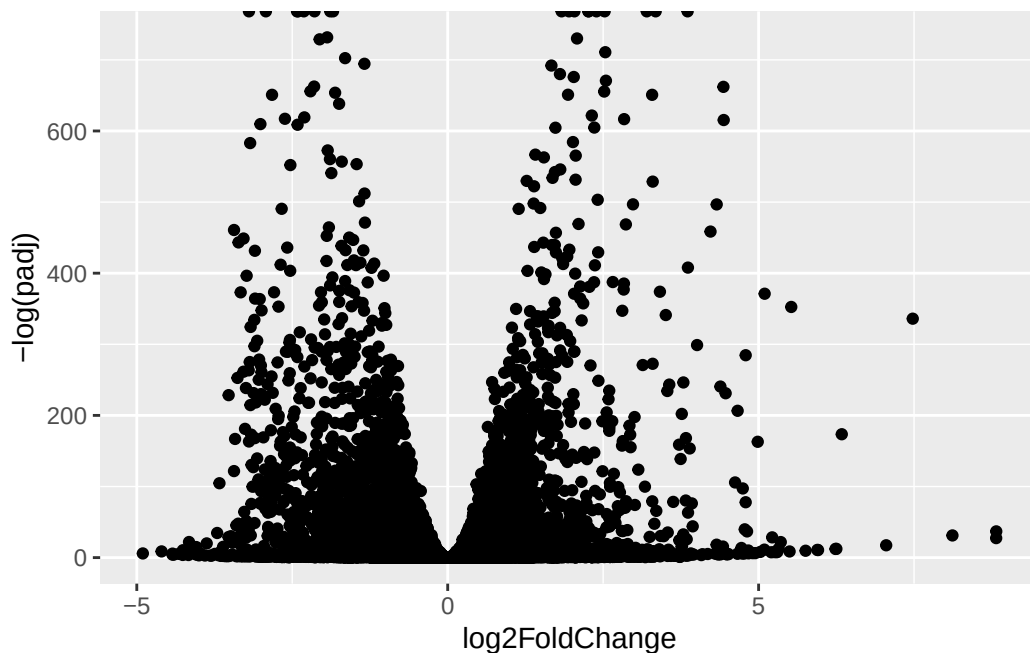
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4393, 27%
outliers [1]      : 0, 0%
low counts [2]    : 1221, 7.6%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

Warning: Removed 5054 rows containing missing values or values outside the scale range (`geom\_point()`).



Improve color, labels, and cutoff:

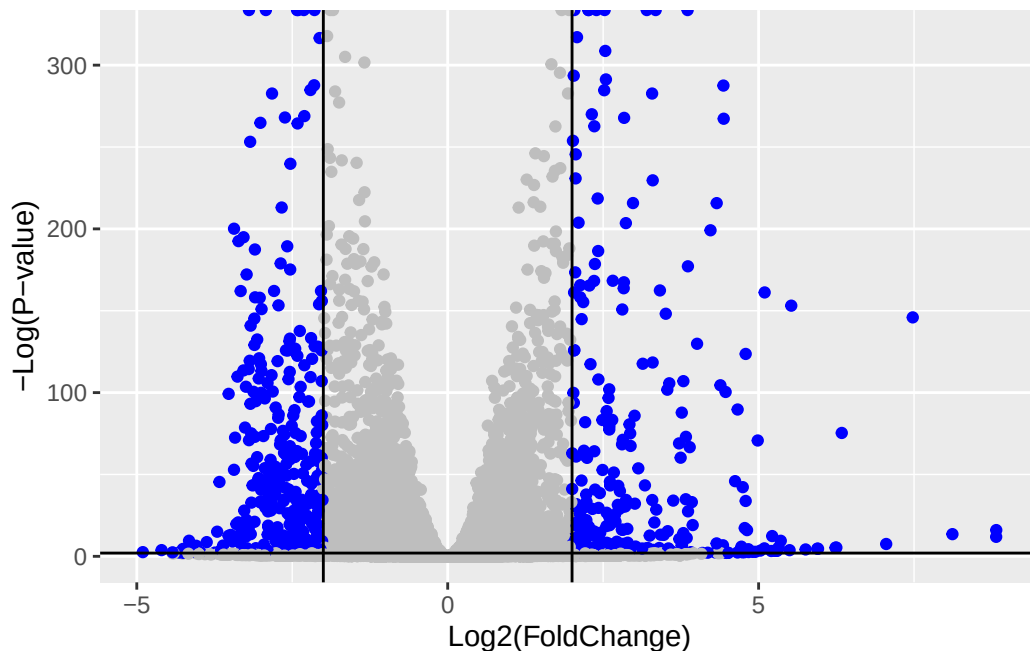
```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(x = log2FoldChange,
      y = -log10(padj)) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log10(0.01))
```

Warning: Removed 5054 rows containing missing values or values outside the scale range (`geom\_point()`).



Gene annotation

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

```
res$symbol = mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="ENTREZID",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
                   keys=row.names(res),
                   keytype="ENSEMBL",
                   column="GENENAME",
                   multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000186092	0.0000	NA	NA	NA	NA
ENSG00000279928	0.0000	NA	NA	NA	NA
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000278566	0.0000	NA	NA	NA	NA
ENSG00000273547	0.0000	NA	NA	NA	NA
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01

	padj	symbol	entrez	name
	<numeric>	<character>	<character>	<character>
ENSG00000186092	NA	OR4F5	79501	olfactory receptor f..
ENSG00000279928	NA	NA	NA	NA
ENSG00000279457	6.87080e-01	NA	NA	NA
ENSG00000278566	NA	NA	NA	NA
ENSG00000273547	NA	NA	NA	NA
ENSG00000187634	5.16278e-03	SAMD11	148398	sterile alpha motif ..
ENSG00000188976	1.76744e-35	NOC2L	26155	NOC2 like nucleolar ..
ENSG00000187961	1.13536e-07	KLHL17	339451	kelch like family me..
ENSG00000187583	9.18988e-01	PLEKHN1	84069	pleckstrin homology ..
ENSG00000187642	4.03817e-01	PERM1	84808	PPARGC1 and ESRR ind..

```
res = res[order(res$pvalue),]
```

```
write.csv(res, file="deseq_results.csv")
```

Pathway Analysis

```
library(pathview)
```

```
#####
```

Pathview is an open source software package distributed under GNU General Public License version 3 (GPLv3). Details of GPLv3 is available at <http://www.gnu.org/licenses/gpl-3.0.html>. Particullary, users are required to formally cite the original Pathview paper (not just mention it) in publications or products. For details, do `citation("pathview")` within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

#####

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
```

```

[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"

```

```

res <- results(dds)
head(res)

```

log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000186092	0.0000	NA	NA	NA	NA
ENSG00000279928	0.0000	NA	NA	NA	NA
ENSG00000279457	29.9136	0.179257	0.324822	0.551863	0.58104203
ENSG00000278566	0.0000	NA	NA	NA	NA
ENSG00000273547	0.0000	NA	NA	NA	NA
ENSG00000187634	183.2296	0.426457	0.140266	3.040350	0.00236304
	padj				
	<numeric>				
ENSG00000186092	NA				
ENSG00000279928	NA				
ENSG00000279457	0.68707975				
ENSG00000278566	NA				
ENSG00000273547	NA				
ENSG00000187634	0.00516278				

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
[1]      NA      NA 0.1792571      NA      NA 0.4264571
```

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names
[1] "greater" "less"      "stats"
```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val	q.val
hsa00232 Caffeine metabolism	NA	NaN	NA	NA
hsa00983 Drug metabolism - other enzymes	NA	NaN	NA	NA
hsa00230 Purine metabolism	NA	NaN	NA	NA
hsa04514 Cell adhesion molecules (CAMs)	NA	NaN	NA	NA
hsa04010 MAPK signaling pathway	NA	NaN	NA	NA
hsa04012 ErbB signaling pathway	NA	NaN	NA	NA

	set.size	expl
hsa00232 Caffeine metabolism	0	NA
hsa00983 Drug metabolism - other enzymes	0	NA
hsa00230 Purine metabolism	0	NA
hsa04514 Cell adhesion molecules (CAMs)	0	NA
hsa04010 MAPK signaling pathway	0	NA
hsa04012 ErbB signaling pathway	0	NA

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa00232" "hsa00983" "hsa00230" "hsa04514" "hsa04010"
```

```
foldchanges <- res$log2FoldChange  
names(foldchanges) <- res$entrez
```

```
foldchanges <- foldchanges[!is.na(names(foldchanges))]  
foldchanges <- foldchanges[!is.na(foldchanges)]
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/eddiecintora/Desktop/BIMM 143/lab14

Info: Writing image file hsa00232.pathview.png

Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/eddiecintora/Desktop/BIMM 143/lab14

Info: Writing image file hsa00983.pathview.png

Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/eddiecintora/Desktop/BIMM 143/lab14

Info: Writing image file hsa00230.pathview.png

Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)

Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/eddiecintora/Desktop/BIMM 143/lab14

Info: Writing image file hsa04514.pathview.png

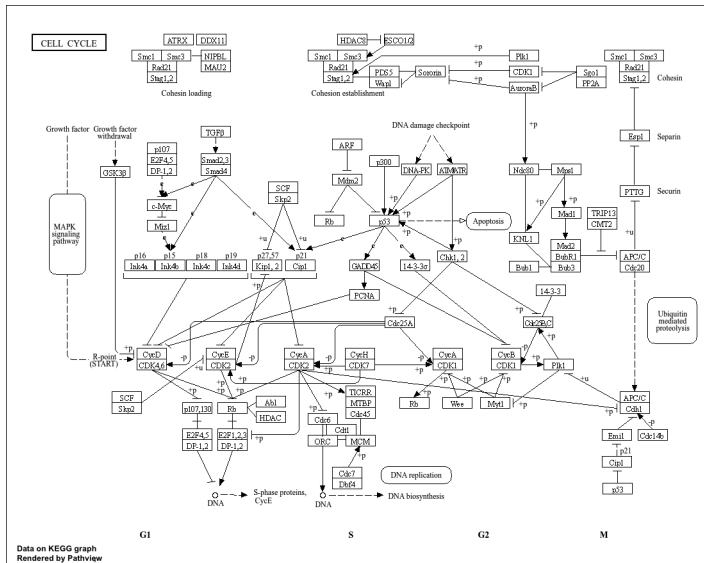
Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/eddiecintora/Desktop/BIMM 143/lab14

Info: Writing image file hsa04010.pathview.png

```
knitr::include_graphics("hsa04110.pathview.png")
```



Gene Ontology

```
data(go.sets.hs)
data(go.subs.hs)
```

```
gobpsets = go.sets.hs[go.subs.hs$BP]
gobpres = gage(foldchanges, gsets=gobpsets)
lapply(gobpres, head)
```

```
$greater
```

		p.geomean	stat.mean	p.val	q.val
G0:0000002	mitochondrial genome maintenance	NA	NaN	NA	NA
G0:0000003	reproduction	NA	NaN	NA	NA
G0:0000012	single strand break repair	NA	NaN	NA	NA
G0:0000018	regulation of DNA recombination	NA	NaN	NA	NA
G0:0000019	regulation of mitotic recombination	NA	NaN	NA	NA
G0:0000022	mitotic spindle elongation	NA	NaN	NA	NA

	set.size	exp1
G0:0000002 mitochondrial genome maintenance	0	NA
G0:0000003 reproduction	0	NA
G0:0000012 single strand break repair	0	NA
G0:0000018 regulation of DNA recombination	0	NA
G0:0000019 regulation of mitotic recombination	0	NA
G0:0000022 mitotic spindle elongation	0	NA

\$less

	p.geomean	stat.mean	p.val	q.val
G0:0000002 mitochondrial genome maintenance	NA	NaN	NA	NA
G0:0000003 reproduction	NA	NaN	NA	NA
G0:0000012 single strand break repair	NA	NaN	NA	NA
G0:0000018 regulation of DNA recombination	NA	NaN	NA	NA
G0:0000019 regulation of mitotic recombination	NA	NaN	NA	NA
G0:0000022 mitotic spindle elongation	NA	NaN	NA	NA

	set.size	exp1
G0:0000002 mitochondrial genome maintenance	0	NA
G0:0000003 reproduction	0	NA
G0:0000012 single strand break repair	0	NA
G0:0000018 regulation of DNA recombination	0	NA
G0:0000019 regulation of mitotic recombination	0	NA
G0:0000022 mitotic spindle elongation	0	NA

\$stats

	stat.mean	exp1
G0:0000002 mitochondrial genome maintenance	NaN	NA
G0:0000003 reproduction	NaN	NA
G0:0000012 single strand break repair	NaN	NA
G0:0000018 regulation of DNA recombination	NaN	NA
G0:0000019 regulation of mitotic recombination	NaN	NA
G0:0000022 mitotic spindle elongation	NaN	NA

```
gogres <- gage(foldchanges,
gsets = go.sets.hs,
ref = NULL,
samp = NULL)
```

Upregulated

```
head(gogres$greater)
```

		p.geomean	stat.mean	p.val	q.val
G0:0000002	mitochondrial genome maintenance	NA	NaN	NA	NA
G0:0000003	reproduction	NA	NaN	NA	NA
G0:0000012	single strand break repair	NA	NaN	NA	NA
G0:0000018	regulation of DNA recombination	NA	NaN	NA	NA
G0:0000019	regulation of mitotic recombination	NA	NaN	NA	NA
G0:0000022	mitotic spindle elongation	NA	NaN	NA	NA

		set.size	exp1
G0:0000002	mitochondrial genome maintenance	0	NA
G0:0000003	reproduction	0	NA
G0:0000012	single strand break repair	0	NA
G0:0000018	regulation of DNA recombination	0	NA
G0:0000019	regulation of mitotic recombination	0	NA
G0:0000022	mitotic spindle elongation	0	NA

Downregulated

```
head(gogres$less)
```

		p.geomean	stat.mean	p.val	q.val
G0:0000002	mitochondrial genome maintenance	NA	NaN	NA	NA
G0:0000003	reproduction	NA	NaN	NA	NA
G0:0000012	single strand break repair	NA	NaN	NA	NA
G0:0000018	regulation of DNA recombination	NA	NaN	NA	NA
G0:0000019	regulation of mitotic recombination	NA	NaN	NA	NA
G0:0000022	mitotic spindle elongation	NA	NaN	NA	NA

		set.size	exp1
G0:0000002	mitochondrial genome maintenance	0	NA
G0:0000003	reproduction	0	NA
G0:0000012	single strand break repair	0	NA
G0:0000018	regulation of DNA recombination	0	NA
G0:0000019	regulation of mitotic recombination	0	NA
G0:0000022	mitotic spindle elongation	0	NA

Reactome Analysis

```
library(AnnotationDbi)
library(org.Hs.eg.db)

res$symbol <- mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENSEMBL",
  column="SYMBOL",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

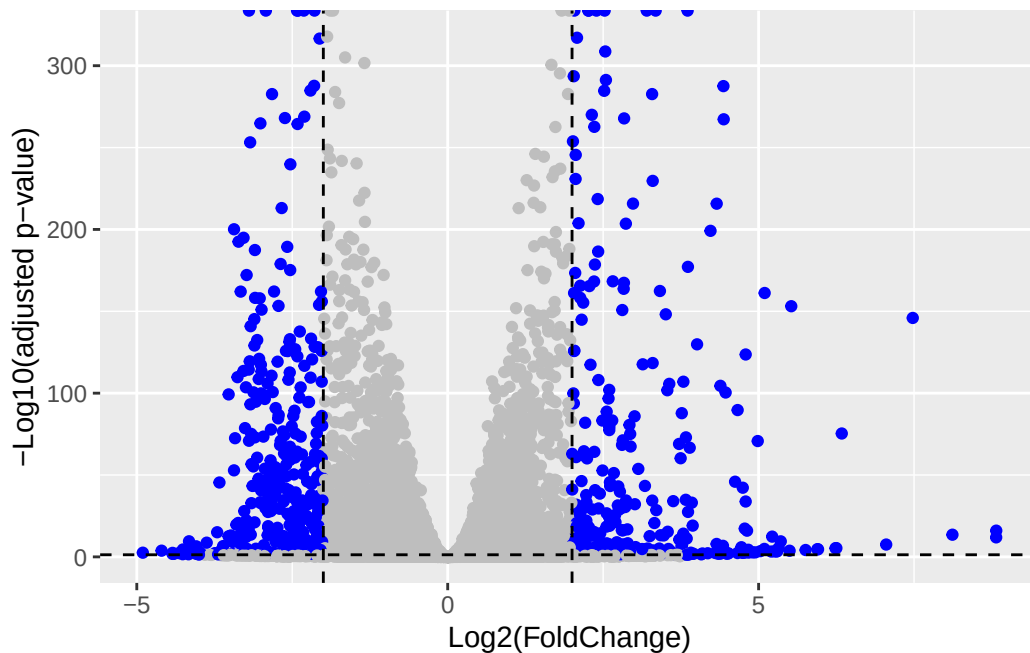
```
library(ggplot2)

mycols <- rep("gray", nrow(res))

mycols[abs(res$log2FoldChange) > 2] <- "blue"
mycols[res$padj > 0.05] <- "gray"

ggplot(res, aes(x = log2FoldChange,
  y = -log10(padj))) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log10(adjusted p-value)") +
  geom_vline(xintercept = c(-2,2), linetype = 2) +
  geom_hline(yintercept = -log10(0.05), linetype = 2)
```

Warning: Removed 5054 rows containing missing values or values outside the scale range (``geom_point()``).



```
sig_genes <- res$symbol[res$padj < 0.05]
```

```
sig_genes <- sig_genes[!is.na(sig_genes)]
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Top Pathway: Cell Cycle, Entities p-value: 2.6e-5

The pathway with the most significant “Entities p-value” was the Cell Cycle pathway. Several of the top Reactome pathways matched the earlier KEGG results, particularly pathways related to the cell cycle and mitosis. Differences between Reactome and KEGG results can occur because the databases use different pathway definitions, gene annotations, and enrichment methods, which can change how genes are grouped and ranked statistically.

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.6.1
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
```

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; 1

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles

tzcode source: internal

attached base packages:

[1] stats4 stats graphics grDevices utils datasets methods
[8] base

other attached packages:

[1] gageData_2.48.0	gage_2.60.0
[3] pathview_1.50.0	org.Hs.eg.db_3.22.0
[5] AnnotationDbi_1.72.0	ggplot2_4.0.3
[7] DESeq2_1.50.2	SummarizedExperiment_1.40.0
[9] Biobase_2.70.0	MatrixGenerics_1.22.0
[11] matrixStats_1.5.0	GenomicRanges_1.62.1
[13] Seqinfo_1.0.0	IRanges_2.44.0
[15] S4Vectors_0.48.1	BiocGenerics_0.56.0
[17] generics_0.1.4	

loaded via a namespace (and not attached):

[1] KEGGREST_1.50.0	gtable_0.3.6	xfun_0.57
[4] lattice_0.22-9	bitops_1.0-9	vctrs_0.7.3
[7] tools_4.5.3	parallel_4.5.3	tibble_3.3.1
[10] RSQLite_3.52.0	blob_1.3.0	pkgconfig_2.0.3
[13] Matrix_1.7-5	RColorBrewer_1.1-3	S7_0.2.2
[16] graph_1.88.1	lifecycle_1.0.5	compiler_4.5.3
[19] farver_2.1.2	Biostrings_2.78.0	codetools_0.2-20
[22] htmltools_0.5.9	RCurl_1.98-1.18	yaml_2.3.12
[25] GO.db_3.22.0	crayon_1.5.3	pillar_1.11.1
[28] BiocParallel_1.44.0	DelayedArray_0.36.1	cachem_1.1.0
[31] abind_1.4-8	tidyselect_1.2.1	locfit_1.5-9.12
[34] digest_0.6.39	dplyr_1.2.1	labeling_0.4.3
[37] fastmap_1.2.0	grid_4.5.3	cli_3.6.6
[40] SparseArray_1.10.10	magrittr_2.0.5	S4Arrays_1.10.1
[43] XML_3.99-0.23	withr_3.0.2	scales_1.4.0
[46] bit64_4.8.0	rmarkdown_2.31	XVector_0.50.0
[49] httr_1.4.8	bit_4.6.0	otel_0.2.0
[52] png_0.1-9	memoise_2.0.1	evaluate_1.0.5
[55] knitr_1.51	rlang_1.2.0	Rcpp_1.1.1-1.1

[58]	glue_1.8.1	DBI_1.3.0	Rgraphviz_2.54.0
[61]	KEGGgraph_1.70.0	rstudioapi_0.18.0	jsonlite_2.0.0
[64]	R6_2.6.1		